



Cairo University
 Faculty of Engineering
 Credit Hours System
Communications and Computer Engineering Department

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Course Code EECS304
 Course Title Control I
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Sheet 9: SSR Canonical Forms

Question (1):

Transform the following transfer function into the observable and controllable canonical forms:

$$\frac{Y(s)}{R(s)} = \frac{4s^2 + 3s + 5}{s^3 + 2s^2 + 4s + 1}$$

Question (2):

Transform the following transfer function into the diagonal form:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 2s + 1}{(s + 1)(s + 0.5)(s + 2)}$$

Question (3):

Given the following state space model:

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -21 & -16 & -6 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u, \quad y = [10 \quad 15 \quad 5] \underline{x}$$

1. Test this model for both controllability and observability.
2. Obtain the transfer function for this system.

Question (4):

A system is described by:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Test this model for both controllability and observability.